Darshan UNIVERSITY

BTech CSE | SEM-5 | Design and Analysis of Algorithm (2301CS402) | 2026-27 (ODD) | Practical List

Lab	Cat	Program
1	A	Write a recursive program for calculation of factorial of an integer.
1	A	Write a program to calculate the sum of numbers from 1 to n using recursion.
1	A	Write a program to count the digits of a given number using recursion.
1	A	Write a program to calculate the power of any number using recursion.
1	B	Write a program to print Fibonacci series for n number using recursion.
1	B	Given a number n, write an efficient function to print all prime factors of n. For example, if the input number is 12, then output should be "2 2 3". And if the input number is 315, then output should be "3 3 5 7".
1	C	Given two integers n and m (m! = 0). Find the number closest to n and divisible by m. If there is more than one such number, then output the one having maximum absolute value. Examples: Input: n = 13, m = 4; Output: 12 Explanation: 12 is the closest to 13, divisible by 4. Input: n = -15, m = 6 ; Output: -18 Explanation: Both -12 and -18 are closest to -15, but -18 has the maximum absolute value. Write a program using two approaches 1. Naive Approach: Iterative Checking - O(m) Time and O(1) Space 2. Expected Approach: By finding Quotient - O(1) Time and O(1) Space
2	A	Write a program to implement stack operations (PUSH, POP, PEEP, CHANGE & DISPLAY)
2	A	Write a program to implement queue operations (INSERT, DELETE, DISPLAY)
2	A	Write a program to implement singly linked list operations (INSERT, DELETE, DISPLAY)
2	B	Given an array arr[] of size n, the task is to rearrange it in alternate positive and negative manner without changing the relative order of positive and negative numbers. In case of extra positive/negative numbers, they appear at the end of the array.
2	B	Given an array arr[], the task is to generate all the possible subarrays of the given array. Examples: Input: arr[] = [1, 2, 3] Output: [[1], [1, 2], [2], [1, 2, 3], [2, 3], [3]] Input: arr[] = [1, 2] Output: [[1], [1, 2], [2]]
2	C	Given a positive integer n, find whether it can be represented as the sum of two or more consecutive positive integers. Examples: Input: n = 10; Output: true Explanation: 10 can be expressed as: 1 + 2 + 3 + 4 = 10. Input: n = 8; Output: false Explanation: 8 cannot be expressed as the sum of two or more consecutive positive integers. Input: n = 24; Output: true Explanation: 24 can be expressed as: 7 + 8 + 9 = 24.
3	A	Write a program to sort array elements using bubble sort. Read n elements given by user from file and observe cpu time taken.
3	A	Write a program to sort array elements using insertion sort. Read n elements given by user from file and observe cpu time taken.
3	A	Write a program to sort array elements using selection sort. Read n elements given by user from file and observe cpu time taken.

3	B	Given an unsorted array, find the minimum difference between any pair in the given array using sorting approad. Examples: Input: [1, 5, 3, 19, 18, 25]; Output: 1 Explanation: Minimum difference is between 18 and 19 Input: [30, 5, 20, 9]; Output: 4 Explanation: Minimum difference is between 5 and 9 Input: [1, 19, -4, 31, 38, 25, 100]; Output: 5 Explanation: Minimum difference is between 1 and -4
3	C	Given a 2D array arr[][] where arr[i][0] represents the starting time and arr[i][1] represents the ending time of the ith meeting, determine whether it is possible for a person to attend all meetings without any overlaps, considering that a person can attend only one meeting at any given time. Input: arr[][] = [[2, 4], [1, 2], [7, 8], [5, 6], [6, 8]] Output: false Explanation: Since the third and fifth meeting overlap, a person cannot attend all the meetings. Input: arr[][] = [[1, 4], [10, 15], [7, 10]] Output: true Explanation: Since all the meetings are held at different times, it is possible to attend all the meetings.
4	A	Write a program to sort array elements using heap sort.
4	B	Given two integer arrays a[] and b[] of the same length, and an positive integer k, the goal is to find the top k maximum sum combinations, where each combination is formed by adding one element from a and one from b using heap sort. Each index from both arrays can be used at most once in a pair. Return the k largest sums in descending order. Examples: Input: a[] = [3, 2], b[] = [1, 4], k = 2 Output: [7, 6] Explanation: Possible sums: 3 + 1 = 4, 3 + 4 = 7, 2 + 1 = 3, 2 + 4 = 6, Top 2 sums are 7 and 6. Input: a[] = [1, 4, 2, 3], b[] = [2, 5, 1, 6], k = 3 Output: [10, 9, 9] Explanation: The top 3 maximum possible sums are : 4 + 6 = 10, 3 + 6 = 9, and 4 + 5 = 9.
5	A	Write a program to implement binary search algorithm using iterative method.
5	A	Write a program to implement binary search algorithm using recursive method.
5	B	Write a program to implement binary search algorithm using linked list and recursive method.
5	C	Given the root of a Binary Search Tree (BST) of size n and an integer target, determine whether there exists a pair of distinct nodes in the BST such that the sum of their values is equal to the given target using in order traversal. Return true if such a pair exists; otherwise, return false. Example: Input: root = [7, 3, 8, 2, 4, N, 9], target = 12 Output: true Explanation: In the given binary tree, there are two nodes (3 and 9) that add up to 12.
6	A	Write a program to implement quick sort algorithm.
6	A	Write a program to implement merge sort algorithm.

6	B	Move all Zeros to End of Array. Given an array of integers arr[], move all the zeros to the end of the array while maintaining the relative order of all non-zero elements. Examples: Input: arr[] = [1, 2, 0, 4, 3, 0, 5, 0]; Output: [1, 2, 4, 3, 5, 0, 0, 0] Explanation: There are three 0s that are moved to the end. Input: arr[] = [10, 20, 30]; Output: [10, 20, 30] Explanation: No change in array as there are no 0s. Input: arr[] = [0, 0]; Output: [0, 0] Explanation: No change in array as there are all 0s.
6	C	Merge Two Sorted Arrays Without Extra Space. Given two sorted arrays a[] and b[] of size n and m respectively, merge both the arrays and rearrange the elements such that the smallest n elements are in a[] and the remaining m elements are in b[]. All elements in a[] and b[] should be in sorted order. Examples: Input: a[] = [2, 4, 7, 10], b[] = [2, 3] ; Output: a[] = [2, 2, 3, 4], b[] = [7, 10] Explanation: Combined sorted array = [2, 2, 3, 4, 7, 10], array a[] contains smallest 4 elements: 2, 2, 3 and 4, and array b[] contains remaining 2 elements: 7, 10. Input: a[] = [1, 5, 9, 10, 15, 20], b[] = [2, 3, 8, 13]; Output: a[] = [1, 2, 3, 5, 8, 9], b[] = [10, 13, 15, 20] Explanation: Combined sorted array = [1, 2, 3, 5, 8, 9, 10, 13, 15, 20], array a[] contains smallest 6 elements: 1, 2, 3, 5, 8 and 9, and array b[] contains remaining 4 elements: 10, 13, 15, 20. Input: a[] = [0, 1], b[] = [2, 3]; Output: a[] = [0, 1], b[] = [2, 3] Explanation: Combined sorted array = [0, 1, 2, 3], array a[] contains smallest 2 elements: 0 and 1, and array b[] contains remaining 2 elements: 2 and 3.
7	A	Write a program to study and implement minimum spanning tree using Kruskal's algorithm.
7	A	Write a program to study and implement minimum spanning tree using Prim's algorithm.
8	A	Write a program to study and implement Dijkstra's algorithm.
8	A	Write a program to implement Knapsack problem using greedy approach.
9	A	Write a program to implement Huffman code algorithm.
9	A	Write a program to implement job scheduling problem using greedy approach.
10	A	Write a program to implement making a change problem using dynamic programming.
10	A	Write a program to Implement 0/1 knapsack problem using dynamic programming.
11	A	Write a program to solve all-pairs shortest paths problem using Floyd's algorithm
11	A	Write a program to implement Largest Common Sub-sequence.
12	A	Write a program to implement the DFS algorithm.
12	A	Write a program to implement the BFS algorithm.
13	A	Write a Program for N Queen's problem using Backtracking.
14	A	Write a program to implement Rabin-Karp method for pattern searching.
15	A	Create a visual application using HTML and JavaScript that demonstrates one of the various algorithms.